Qihuang SmartHand

TCM Digital Acupressure & Acupoint Quantitative Assessment System

Zhaoqing Medical College | Guangdong Maker Competition (Entrepreneurship Group)

1. Project Overview

Project Name:	Qihuang SmartHand (QiHuang ZhiShou)
Full Name:	TCM Digital Acupressure Technique & Acupoint Quantitative Assessment System
Institution:	Zhaoqing Medical College
Competition:	Maker Guangdong 2026 - Entrepreneurship Group (ChuangKe Team)
Stage:	Preliminary Round (8 min pitch + 4 min Q&A)
Field:	Medical Health / Digital TCM / Medical Devices

2. Problem & Market Pain Points

Traditional Chinese Medicine (TCM) acupressure and acupuncture training faces three critical bottlenecks: (1) Lack of quantitative standards - acupoint location relies entirely on master-apprentice oral instruction, with no digital force/angle/depth data; (2) Low training efficiency - a TCM practitioner requires 8-10 years of clinical experience to achieve consistent acupoint accuracy; (3) No objective assessment system - existing exams and skill certifications are entirely subjective, with zero quantitative metrics for acupoint manipulation quality. The TCM education market in China is valued at RMB 120 billion (TAM), with over 600,000 TCM students and 2.8 million practicing TCM physicians who lack standardized skill assessment tools.

3. Solution & Core Technology

Qihuang SmartHand is a hardware-software integrated system combining flexible pressure sensor arrays, 6-axis IMU motion capture, and a DTW (Dynamic Time Warping) algorithm engine to digitize the entire acupressure manipulation process. The system captures real-time force distribution (0.1N resolution), finger posture (pitch/yaw/roll), pressing depth (0.1mm accuracy), and rhythm patterns, then matches them against a library of 361 standard acupoint manipulation templates using our DTW-based scoring engine.

Core Technology Stack:

- Flexible Thin-Film Pressure Sensor Array (64-point matrix, 0-50N range)
- 6-Axis IMU Motion Capture (0.01-degree angular resolution)
- DTW Algorithm Engine: acupoint matching accuracy 95% (validated on 2,000+ clinical samples)
- Proprietary TCM Acupoint Database: 361 acupoints x 5 standard manipulation parameters
- Real-time AI Feedback: latency < 200ms from sensor to visual/audio feedback
- Cloud + Edge Architecture: training data sync, performance analytics dashboard

4. Market Analysis & Competition

Total Addressable Market:	RMB 120 billion (TCM education & assessment)
Serviceable Market (SAM):	RMB 18 billion (TCM universities + teaching hospitals)
Target Market (SOM):	RMB 360 million (Year 1-3, 60 pilot institutions)

Competitive Landscape:

- Direct Competitors: None. No existing product combines flexible sensor arrays + DTW for TCM acupoint quantitative assessment.
- Indirect Competitors: (a) Traditional TCM training mannequins - no digital feedback; (b) General medical simulation (CAE Healthcare) - no TCM specialization; (c) Wearable IMU gloves (Rokoko, Manus) - no pressure sensing or TCM database.
- Moat: Proprietary TCM acupoint manipulation database + DTW scoring algorithm + 3 invention patents filed.

5. Business Model

Product Pricing:	RMB 3,980 per unit (hardware sensor glove + software license)
Annual SaaS:	RMB 1,200 per student per year (cloud analytics + database updates)
Revenue Streams:	Hardware sales (60%), SaaS subscriptions (25%), Custom database licensing (15%)
Target Customers:	TCM universities (30+), TCM teaching hospitals (2,000+), TCM skill certification bodies

6. Financial Projections (3-Year)

Year 1 Revenue:	RMB 2.4 million (600 units x RMB 3,980)
Year 2 Revenue:	RMB 8.5 million (2,000 units + SaaS recurring)
Year 3 Revenue:	RMB 22 million (5,000 units + SaaS + database licensing)
IRR (5-year):	52%
NPV (5-year, WACC 10%):	RMB 5.8 million
Breakeven:	Month 14

7. Funding Request

Seeking:	RMB 900,000 Angel Round
Equity Offered:	10%
Pre-money Valuation:	RMB 8.1 million
Use of Funds:	40% Hardware R&D (sensor optimization + mold), 30% Clinical validation (3 hospitals), 20% Software development (cloud platform), 10% IP & certification

8. Intellectual Property

- Invention Patent (filed): Flexible Acupoint Pressure Sensing Apparatus and Method (CN2025XXXXXX)
- Invention Patent (filed): DTW-Based Acupoint Manipulation Scoring System (CN2025XXXXXX)
- Invention Patent (filed): Multi-Modal TCM Acupoint Training Feedback System (CN2025XXXXXX)
- Software Copyright: Qihuang SmartHand Assessment Platform V1.0
- Trademark: Qihuang SmartHand (QiHuang ZhiShou) - Class 9, 10, 42

9. Core Team

Team Lead:	Guan Ziyue - Zhaoqing Medical College, TCM clinical background, project management
Tech Lead:	Hardware/embedded systems engineer, sensor integration experience

Algorithm Lead:	ML/DSP specialist, DTW and time-series analysis expertise
Clinical Advisor:	Associate Chief TCM Physician, 20+ years acupressure clinical experience
Business Lead:	MedTech startup experience, medical device sales channel background
Team Size:	7 core members (cross-disciplinary: TCM + Engineering + CS + Business)

10. Current Status & Roadmap

Completed:

- Working prototype v2.0 (flexible sensor glove + IMU module)
- 361-acupoint database with 5 standard manipulation parameters each
- DTW scoring engine: 95% accuracy on 2,000 clinical samples
- 3 invention patents filed, 1 software copyright registered

Next 6 Months:

- Clinical validation trial at 3 TCM teaching hospitals (n=300)
- Cloud analytics platform MVP launch
- Medical device registration (Class II) preparation
- Seed round fundraising close

11. Key Metrics at a Glance

DTW Accuracy:	95%
Sensor Resolution:	0.1N force, 0.1mm depth, 0.01-degree angle
Acupoint Database:	361 acupoints x 5 parameters
Response Latency:	< 200ms
Unit Price:	RMB 3,980
TAM:	RMB 120 billion
IRR (5yr):	52%
NPV (5yr):	RMB 5.8M
Funding Ask:	RMB 900K / 10% equity
Patents:	3 invention patents filed